



P 4

S y s t e m

Database Management System Mini Project — a fully functional relational database for managing hotel operations including room booking, customer management, and billing.

MICROSOFT SQL SERVER

T-SQL

SSMS

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P r o b l e m

Design a robust, normalized database system to manage end-to-end hotel operations — from guest check-in to billing — with full data integrity and automation.

S c a l a b l e

Normalized up to BCNF for clean, efficient data organization

D a t a

Enforced via PRIMARY KEY, FOREIGN KEY, UNIQUE, and CHECK constraints

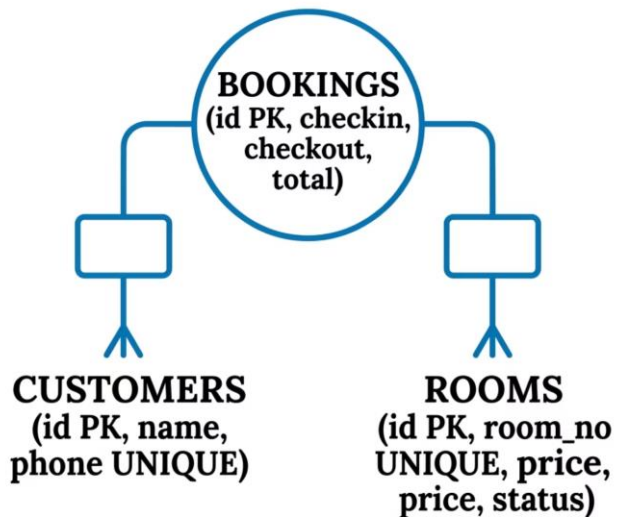
A u t o m a t i o n

Triggers handle total calculation and room status updates automatically

S e c u r i t y

Transaction control (TCL) and role-based access (DCL) fully implemented

D a t a b a s e



T h r e e

- **Customers**— id (PK), name, phone (UNIQUE)
- **Rooms**— id (PK), room_no (UNIQUE), price, status
- **Bookings**— id (PK), cust_id (FK), room_id (FK), checkin, checkout, total

R e l a t i o n s h i p s

- Customers → Bookings: **One-to-Many**
- Rooms → Bookings: **One-to-Many**

③ Normalization achieved up to **BCNF (Boyce-Codd Normal Form)**, eliminating all redundancy and update anomalies.

Feature

Core

D D L

CREATE, DROP, ALTER

C o n s t r a i n t s

PK, FK, UNIQUE,
NOT NULL, CHECK

D M L

INSERT, UPDATE, DELETE

I n d e x e s

Optimized query performance

Advanced

V i e w s

vw_AvailableRooms, vw_AllBookings

T r i g g e r s

trg_CalcTotal, trg_UpdateRoom

T C L

COMMIT, ROLLBACK, GRANT, REVOKE

F u n c t i o n s

fn_Nights — calculates stay duration

J o i n s

INNER JOIN, LEFT JOIN, GROUP BY,
HAVING

7

S Q L

3

T a b l e s

3

T o t a l

2

T r i g g e r s

Query

1

I N N J E O R I - N C u s t o m e r

Joins Customers → Bookings → Rooms to display guest names with assigned room numbers.

Output: Raj → Room 101 | Priya → Room 201

2

V I E-WA v a i l a b l e

Queries `vw AvailableRooms` to list all unbooked rooms with pricing.

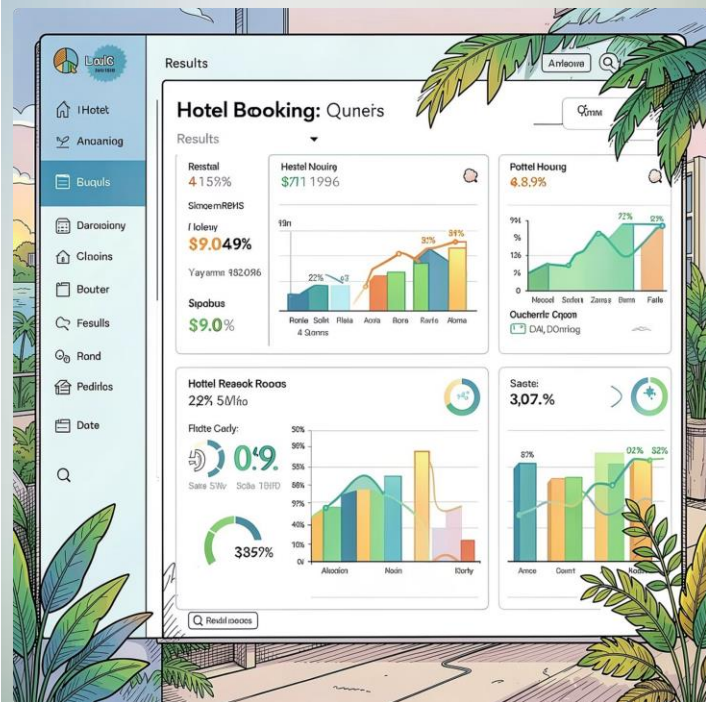
Output: Rooms 101, 102, 201, 202, 301 with prices ₹3000-₹8000

3

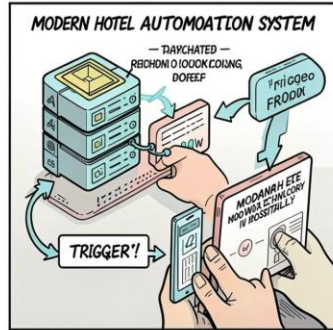
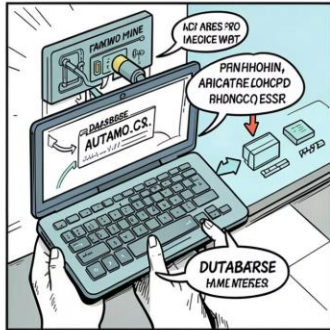
S U B Q U E R E d

Uses a nested SELECT to retrieve names of all customers with active bookings.

Output: Raj, Priya, Amit



Functions



fn_Nights

Inputs: check-in date, check-out date · **Returns:** number of nights
Example: `fn_Nights('2026-06-01', '2026-06-03')` → 2 nights

trg_CalculateTotalTrigger

Fires on Bookings insert. Auto-calculates `total = Nights × Room Price`, eliminating manual entry errors.

trg_UpdateRoomStatus

Fires after booking confirmation. Updates room status from 'Available' to 'Booked', preventing double bookings.



Automation Flow: Booking Insert → Calculate Total → Update Room Status → Complete

T r a n s a c t i o n

T r a n s a c t i o n

Ensures data consistency through atomic operations — either all changes commit or none do.

D a t a

Role-based access control restricts user permissions to protect sensitive data.

```
CREATE USER hotel_user WITHOUT LOGIN;  
GRANT SELECT ON vw_AvailableRooms TO hotel_user;  
REVOKE INSERT ON Customers FROM hotel_user;
```

 **Result:** hotel_user has read-only access to room reports — no insert or delete permissions on core tables.

C O M M I T

```
BEGIN TRANSACTION; INSERT INTO Bookings...; COMMIT;
```

Data permanently saved to database.

R O L L B A C K

```
BEGIN TRANSACTION; DELETE FROM Bookings...; ROLLBACK;
```

Deletion cancelled, data fully restored.

S u m m a r y

✓ W h a t



Database normalized to **BCNF** with 7 tested SQL scripts



Full **DDL, DML, DCL, TCL** implementation



Joins, Subqueries, and Views operational



2 automated Triggers + 1 scalar Function



Index created for query performance

🚀 F u t u r e



P a y m e n t

Online payment integration for seamless checkout



L i v e

Real-time room availability and occupancy analytics



M o b i l e

Customer-facing app for booking and check-in



L o y a l t y

Reward points and repeat-guest benefits

Project Repository: github.com/PrashastiSrivastava/